

Question Number	Answer	Additional Guidance	Mark
4(a)	A description that includes the following points: • suitable way of germinating seeds/checking they are viable (1) • find a suitable temperature for {respiration/germination/seeds to grow} (1) • find a suitable mass of seeds to give a measurable volume of gas(1) • find suitable method for absorbing carbon dioxide (1) • find a suitable method to measure (change of) gas volume (1)	Do not allow oxygen <u>produced</u>	(2)

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4(b)	An answer that includes the following points: • clear statement of the dependent variable e.g. distance moved in unit time volume of oxygen in unit time (1) • some description of apparatus used (1) • control mass of seeds (1) • (record) time for a measured distance of the meniscus or volume of gas (1) • time to acclimatise (1)	Do not piece together the dependent variable Allow volume of gas using syringe allow respirometer/labelled diagram/manometer allow KOH/NaOH	(9)

	• repeat with and without soda lime (1) • one variable that needs to be controlled (1) • description of how this variable is controlled (1) • repeat the method with the other seed type (1) • formula for calculating RQ (1)	allow disinfection with bleach to prevent contamination Ignore pH/light/ Allow AC with suitable stated temperature/incubator allow $\text{CO}_2 \div \text{O}_2$ Or distance with soda lime - distance without $\div$ distance without soda lime	
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Question Number	Answer	Additional Guidance	Mark
4(c)	• table with headings for raw results and appropriate units means calculated from repeats (1) • bar graph format with labelled axes (1) • use of an appropriate statistical test (1)	t test/Mann-Whitney U test/Wilcoxon test	(3)

Question Number	Answer	Additional Guidance	Mark
4(d)	• difficult to measure distances or (collect small) volumes of gas (1) • difficult to prevent contamination of watermelon seeds (1) • difficulty of controlling temperature (1)	allow apparatus contaminated ignore temperature in a list of variables	(2)

Total for question 4 = 16

